

Left Lower Quadrant Pain
EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
1. Cartwright SL, Knudson MP. Diagnostic imaging of acute abdominal pain in adults. [Review]. American Family Physician. 91(7):452-9, 2015 Apr 01.	Review/Other-Dx	N/A	To review clinical guidelines that help physicians choose the most appropriate imaging study based on location of abdominal pain.	No results stated in abstract.	4
2. Barat M, Paisant A, Calame P, et al. Unenhanced CT for clinical triage of elderly patients presenting to the emergency department with acute abdominal pain. Diagn Interv Imaging. 100(11):709-719, 2019 Nov.	Observational-Dx	208 patients	To compare the diagnostic accuracy and inter-reader agreement of unenhanced computed tomography (CT) to those of contrast-enhanced CT for triage of patients older than 75 years admitted to emergency department (ED) with acute abdominal pain (AAP).	Diagnostic accuracy ranged from 64% (95% CI: 62-66%) to 68% (95% CI: 66-70%) for unenhanced CT, and from 68% (95% CI: 66-70%) to 71% (95% CI: 69-73%) for both unenhanced and contrast-enhanced CT. Contrast-enhanced CT did not significantly improve the diagnostic accuracy (P=0.973-0.979). CT corrected the working diagnosis proposed by the ED physician in 59.1% (range: 58.1-60.0%) and 61.2% (range: 57.6-65.5%) of patients before and after contrast injection (P>0.05). Intra-observer agreement was moderate to substantial (k=0.513-0.711). Inter-reader agreement was substantial for unenhanced (kappa=0.745-0.789) and combined unenhanced and contrast-enhanced CT (kappa=0.745-0.799). Results were similar in subgroup analyses.	3
3. Bolkenstein HE, van Dijk ST, Consten ECJ, et al. Conservative Treatment in Diverticulitis Patients with Pericolic Extraluminal Air and the Role of Antibiotic Treatment. J Gastrointest Surg. 23(11):2269-2276, 2019 11.	Review/Other-Dx	109 patients	To assess the clinical course of initially conservatively treated diverticulitis patients with isolated pericolic air and to identify risk factors for conservative treatment failure.	Fifty-two (48%) patients were treated with antibiotics. Nine (8%) patients failed conservative management, seven (13%) in the antibiotic treatment group and two (4%) in the non-antibiotic group (p = 0.083). Only (increased) CRP level at presentation was an independent predictor for treatment failure.	4
4. Caputo P, Rovagnati M, Carzaniga PL. Is it possible to limit the use of CT scanning in acute diverticular disease without compromising outcomes? A preliminary experience. Ann Ital Chir. 86(1):51-5, 2015 Jan-Feb.	Observational-Dx	93 patients	To determine whether the use of CT scanning in the assessment of acute diverticulitis can be reduced without a negative effect on outcome.	In patients with a HS ≤ 1b surgical intervention was never required and therefore we believe that US imaging is sufficient and they can be spared the exposure to ionizing radiation associated with CT scans.	4

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5. Juszczuk K, Ireland K, Thomas B, Kroon HM, Hollington P. Reduction in hospital admissions with an early computed tomography scan: results of an outpatient management protocol for uncomplicated acute diverticulitis. ANZ J Surg. 89(9):1085-1090, 2019 09.	Observational-Dx	1147 patients	To evaluate if an early computed tomography (CT) scan in the emergency department (ED) can reduce the number of hospital admissions when UAD is diagnosed, without compromising patient safety.	There was a significant decrease in hospital admissions for UAD in the post-protocol group from 93% to 39% ($P < 0.0001$) and in the total number of hospital admission days from 602 to 370 ($P < 0.0001$). There was no increase in representations between both periods (7% versus 6%; $P = 0.49$).	4
6. Weinrich JM, Bannas P, Avanesov M, et al. MDCT in the Setting of Suspected Colonic Diverticulitis: Prevalence and Diagnostic Yield for Diverticulitis and Alternative Diagnoses. AJR. American Journal of Roentgenology. 215(1):39-49, 2020 07. AJR Am J Roentgenol. 215(1):39-49, 2020 07.	Observational-Dx	1069 patients	To determine the prevalence and demographic distribution of colonic diverticulitis (CD) and alternative diagnoses (AD), as well as the diagnostic accuracy of MDCT in patients with suspected CD.	Prevalence of CD was 52.5% (561/1069) and of AD was 39.9% (427/1069). In the remaining 7.6% (81/1069) no final clinical diagnosis was established. The most frequent AD were appendicitis (12.6%, 54/427), infectious colitis (10.5%, 45/427), infectious gastroenteritis (8.2%, 35/427), urolithiasis (6.1%, 26/427), and pyelonephritis (4.9%, 21/427). The prevalence of diverticulitis and AD varied statistically significantly according to both age ($p < 0.001$) and admission status ($p < 0.001$). Also, the prevalence of the 10 most frequent specific AD varied statistically significantly according to sex ($p = 0.022$). CT had a sensitivity and specificity of 99.1% and 99.8% for diagnosing CD and 92.7% and 98.8% for AD, respectively.	4

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7. Bharucha AE, Parthasarathy G, Ditah I, et al. Temporal Trends in the Incidence and Natural History of Diverticulitis: A Population-Based Study. <i>Am J Gastroenterol</i> 2015;110:1589-96.	Review/Other-Dx	N/A	To assess temporal trends in the epidemiology of diverticulitis in the general population.	In 1980-1989, the incidence of diverticulitis was 115/100,000 person-years, which increased to 188/100,000 in 2000-2007 ($P<0.001$). Incidence increased with age ($P<0.001$); however, the temporal increase was greater in younger people ($P<0.001$). Ten years after the index and second diverticulitis episodes, 22% and 55% had a recurrence, respectively. This recurrence rate was greater in younger people (hazard ratio (HR) per decade 0.63; 95% confidence interval (CI), 0.59-0.66) and women (HR 0.68; 95% CI, 0.58-0.80). Complications were seen in 12%; this rate did not change over time. Recurrent diverticulitis was associated with a decreased risk of complications ($P<0.001$). Age was associated with increased risk of local (odds ratio (OR) 1.27 per decade; 95% CI, 1.04-1.57) and systemic (OR 1.83; 95% CI, 1.20-2.80) complications. Survival after diverticulitis was lower in older people ($P<0.001$) and men ($P<0.001$) and worsened over time ($P<0.001$). The incidence of surgery for diverticulitis did not change from 1980 to 2007.	4
8. Sartelli M, Moore FA, Ansaloni L, et al. A proposal for a CT driven classification of left colon acute diverticulitis. <i>World J Emerg Surg</i> 2015;10:3.	Review/Other-Dx	N/A	To present a new simple classification system based on both CT scan results driving decisions making management of acute diverticulitis that may be universally accepted for day to day practice.	No results stated in abstract.	4
9. Bodmer NA, Thakrar KH. Evaluating the Patient with Left Lower Quadrant Abdominal Pain. [Review]. <i>Radiol Clin North Am.</i> 53(6):1171-88, 2015 Nov.	Review/Other-Dx	N/A	To review the evaluation of a patient with left lower quadrant abdominal pain.	No results stated in abstract.	4

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10. Sai VF, Velayos F, Neuhaus J, Westphalen AC. Colonoscopy after CT diagnosis of diverticulitis to exclude colon cancer: a systematic literature review. <i>Radiology</i> . 2012;263(2):383-390.	Review/Other-Dx	10 articles	To estimate the prevalence of underlying adenocarcinoma of the colon in patients in whom acute diverticulitis was diagnosed at CT and to compare that to the prevalence of colon cancer in the general population.	10 articles met the inclusion criteria. Data from these articles included only 771 patients who underwent surgery, colonoscopy, or BE study within 24 weeks of diagnosis. 14 patients were found to have colon cancer, for a prevalence of 2.1% (95% CI: 1.2%, 3.2%). This compares to a calculated estimated prevalence of 0.68% among U.S. adults >55 years.	4
11. van Dijk ST, Daniels L, Nio CY, Somers I, van Geloven AAW, Boermeester MA. Predictive factors on CT imaging for progression of uncomplicated into complicated acute diverticulitis. <i>Int J Colorectal Dis</i> . 32(12):1693-1698, 2017 Dec.	Observational-Dx	528 patients	To identify computed tomography (CT) imaging predictors for a complicated disease course of initially uncomplicated acute diverticulitis.	Of the 528 patients in the DIABOLO trial, 16 patients developed complications (abscess > 5 cm, perforation, bowel obstruction) within 90 days after randomization. In the group with a complicated course of initially uncomplicated diverticulitis, more patients with fluid collections (25 vs. 0%; p = 0.009) and a longer inflamed colon segment (86 +/- 26 mm vs. 65 +/- 21 mm; p = 0.007) were observed compared to an uncomplicated course of disease. Pericolic extraluminal air was no predictive factor.	3
12. Oistamo E, Hjern F, Blomqvist L, Von Heijne A, Abraham-Nordling M. Cancer and diverticulitis of the sigmoid colon. Differentiation with computed tomography versus magnetic resonance imaging: preliminary experiences. <i>Acta Radiol</i> . 2013;54(3):237-241.	Observational-Dx	30 patients	To determine whether MRI could be helpful to differentiate between diverticulitis and cancer of the sigmoid colon compared to the differentiation offered by evaluation of MDCT in a clinical situation.	With contrast-enhanced CT, the sensitivity and specificity for diagnosis of cancer and diverticulitis were 66.7% (10/15) and 93.3% (14/15), respectively. Using T2-weighted and diffusion-weighted MRIs, the sensitivity and specificity for diagnosis of cancer and diverticulitis were 100% (14/14) and 100% (14/14), respectively.	4

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13. Lahat A, Yanai H, Menachem Y, Avidan B, Bar-Meir S. The feasibility and risk of early colonoscopy in acute diverticulitis: a prospective controlled study. <i>Endoscopy</i> 2007;39:521-4.	Observational-Dx	154 patients	To compare early and late colonoscopy in hospitalized patients with acute diverticulitis.	45 patients were randomly allocated for early colonoscopy and 41 for late colonoscopy. Three and 10 did not present for the examination, in the early and late group respectively. The cecum could not be reached in eight and three patients from the early and late groups, respectively. The colonoscopy revealed polyps in five patients, two in the early group and three in the late group. No malignancy was detected. There were no complications in either group.	3
14. Suhardja TS, Norhadi S, Seah EZ, Rodgers-Wilson S. Is early colonoscopy after CT-diagnosed diverticulitis still necessary?. <i>Int J Colorectal Dis.</i> 32(4):485-489, 2017 Apr.	Observational-Dx	523 patients	To review the current practice by looking at our institution's rate of colorectal malignancy diagnosed after an episode of acute diverticulitis.	A total of 523 cases of acute diverticulitis were diagnosed on CT scan. Out of 351 patients with uncomplicated diverticulitis, 196 had follow-up colonoscopy, with one case of colorectal malignancy recorded. Low-grade and advanced adenomas were found on 10.7 and 2.0% of colonoscopies performed respectively in this subgroup. Seventy-four out of 172 patients with complicated diverticulitis had follow-up evaluation, with four cases of colorectal malignancy discovered. Low-grade and advanced adenomas were found on 6.75 and 5.41% of colonoscopies performed respectively in this subgroup.	4
15. Ou G, Rosenfeld G, Brown J, et al. Colonoscopy after CT-diagnosed acute diverticulitis: Is it really necessary?. <i>Can J Surg.</i> 58(4):226-31, 2015 Aug.	Observational-Dx	293 patients	To determine the prevalence of CRC among patients with CT-diagnosed acute diverticulitis.	A total of 293 patients had acute diverticulitis diagnosed on CT scan, but 8 were nonresidents and were excluded. Of the 285 included in the analysis, the mean age was 59.4 +/- 15.1 years, and 167 (58.6%) were men. Among the 114 patients who underwent follow-up evaluation, malignancy was diagnosed in 4 (3.5%). The overall prevalence of malignancy among patients with CT-diagnosed diverticulitis was 1.4%.	3

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16. Stimac D, Nardone G, Mazzari A, et al. What's New in Diagnosing Diverticular Disease. J. Gastrointestinal Liver Diseases. 28(suppl. 4):17-22, 2019 Dec 19.	Review/Other-Dx	N/A	To consider different issues for the diagnosis of diverticular disease (DD) were considered including "Biomarkers", "Computer tomography", "Ultrasonography in detecting acute diverticulitis", "Endoscopy" and "The DICA classification: a new predictive tool in managing diverticular disease"	No results stated in abstract.	4
17. Kessner R, Barnes S, Halpern P, Makrin V, Blachar A. CT for Acute Nontraumatic Abdominal Pain-Is Oral Contrast Really Required?. Acad Radiol. 24(7):840-845, 2017 07.	Observational-Dx	348 patients	To compare the diagnostic performance of abdominal computed tomography (CT) performed with and without oral contrast in patients presenting to the emergency department (ED) with acute nontraumatic abdominal pain.	Each group consisted of 82 men and 92 women. The average age of the two groups was 48 years. The main clinical diagnoses of the pathological examinations were appendicitis (17.5%), diverticulitis (10.9%), and colitis (5.2%). A normal CT examination was found in 34.8% of the patients. There was no significant difference between the groups regarding most of the clinical parameters that were examined. None of the examinations of all of the 174 study group patients was found to be technically inadequate, and therefore no patient had to undergo additional scanning to establish a diagnosis. The consensus reading of the senior radiologists determined that the lack of oral contrast was insignificant in 96.6% of the cases and that contrast material might have been useful in only 6 of 174 study group patients (3.4%). The radiologists found oral contrast to be helpful only in 8 of 174 control group patients (4.6%). There was no significant difference between the clinical and radiological diagnoses in both groups (study group, P = 0.261; control group, P = 0.075).	3

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18. American College of Radiology. ACR Appropriateness Criteria®: Acute Pelvic Pain in the Reproductive Age Group. Available at: https://acsearch.acr.org/docs/69503/Narrative/ .	Review/Other-Dx	N/A	Evidence-based guidelines to assist referring physicians and other providers in making the most appropriate imaging or treatment decision for a specific clinical condition.	No abstract available.	4
19. Fukata K, Takamizawa J, Miyake H, et al. Diagnosis of appendiceal diverticulitis by multidetector computed tomography. Jpn J Radiol. 38(6):572-578, 2020 Jun.	Observational-Dx	388 patients	To reveal the characteristics of computed tomography (CT) images of AD.	Univariate analysis showed that maximal diameter of the appendix in AD was significantly smaller than that in AA. Multivariate analysis showed that saccular structure of the appendix wall, cecum or ascending colon diverticulum and peri-appendiceal or -cecal fluid collection were significant independent indicators of AD. Based on that saccular structure of the appendix wall which was the most important specific finding, the sensitivity and specificity of CT for AD were 48% and 99%, respectively.	3
20. van Randen A, Lameris W, van Es HW, et al. A comparison of the accuracy of ultrasound and computed tomography in common diagnoses causing acute abdominal pain. Eur Radiol. 2011;21(7):1535-1545.	Observational-Dx	1,021 patients	To report a head-to-head comparison of the accuracy of US and CT in detecting common causes of acute abdominal pain, such as appendicitis and diverticulitis, in patients presenting at the emergency department with acute abdominal pain.	Frequent final diagnoses in the 1,021 patients (mean age 47; 55% female) were appendicitis (284; 28%), diverticulitis (118; 12%) and cholecystitis (52; 5%). The sensitivity of CT in detecting appendicitis and diverticulitis was significantly higher than that of US: 94% vs 76% (P<0.01) and 81% vs 61% (P=0.048), respectively. For cholecystitis, the sensitivity of both was 73% (P=1.00). PPVs did not differ significantly between US and CT for these conditions. US sensitivity in detecting appendicitis and diverticulitis was not significantly negatively affected by patient characteristics or reader experience.	2

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21. Faggian A, Berritto D, Iacobellis F, Reginelli A, Cappabianca S, Grassi R. Imaging Patients With Alimentary Tract Perforation: Literature Review. [Review]. Seminars in Ultrasound, CT & MR. 37(1):66-9, 2016 Feb.Semin Ultrasound CT MR. 37(1):66-9, 2016 Feb.	Review/Other-Dx	N/A	To discuss imaging in the diagnosis of alimentary tract perforation.	No results stated in abstract.	4
22. Tau N, Cohen I, Barash Y, Klang E. Free abdominal gas on computed tomography in the emergency department: aetiologies and association between amount of free gas and mortality. Ann R Coll Surg Engl. 102(8):581-589, 2020 Oct.	Observational-Dx	372 patients	To describe the sources of free abdominal gas on emergency department (ED) computed tomography (CT).	Colonic diverticulitis was the most common aetiology among those with a small or medium amount of free gas (81/250 [32.4%] and 12/71 [16.9%] respectively). For patients with a large amount of gas, peptic disease was the most common aetiology (11/51 [21.6%]). Three-quarters of the patients (280/372, 75.2%) had the source of free gas identified during ED admission. Ninety-day mortality rates were 7.2%, 9.9% and 21.6% for patients with small, medium and large amounts of gas respectively (p=0.007).	3
23. Millet I, Sebbane M, Molinari N, et al. Systematic unenhanced CT for acute abdominal symptoms in the elderly patients improves both emergency department diagnosis and prompt clinical management. European Radiology. 27(2):868-877, 2017 Feb.Eur Radiol. 27(2):868-877, 2017 Feb.	Observational-Dx	401 patients	To assess the added-value of systematic unenhanced abdominal computed tomography (CT) on emergency department (ED) diagnosis and management accuracy compared to current practice, in elderly patients with non-traumatic acute abdominal symptoms.	Systematic unenhanced CT significantly improved the accurate diagnosis (76.8% to 85%, p=1.1x10 ⁻⁶) and management (88.5% to 95.8%, p=2.6x10 ⁻⁶) rates compared to current practice. It allowed diagnosing 30.3% of acute unsuspected pathologies, 3.4% of which were unexpected surgical procedure requirement.	3

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24. Sala E, Watson CJ, Beadmoore C, et al. A randomized, controlled trial of routine early abdominal computed tomography in patients presenting with non-specific acute abdominal pain. Clin Radiol. 2007;62(10):961-969.	Experimental-Dx	198 patients (99 in each arm)	Prospective randomized trial to compare the effect of an initial early CT examination vs standard practice on the length of hospital stay, diagnostic accuracy, and mortality of adults presenting with acute abdominal pain.	There was no significant difference in the length of hospital stay between the two arms (P=0.20). At randomization 36% (35/96) of CT patients and 49% (48/98) of standard practice patients were correctly diagnosed; 24 hours after randomization the correct diagnosis had been established in 84% of CT patients and 73% of standard practice patients. This refinement in diagnostic certainty was significantly better in the CT group (P<0.001). There was no difference in mortality between the two trial arms (P=0.31). Early abdominal CT in patients with acute abdominal pain improves diagnostic certainty, but does not reduce the length of hospital stay and 6 month mortality.	2
25. Dirks K, Calabrese E, Dietrich CF, et al. EFSUMB Position Paper: Recommendations for Gastrointestinal Ultrasound (GIUS) in Acute Appendicitis and Diverticulitis. [Review]. Ultraschall in der Medizin. 40(2):163-175, 2019 Apr. Ultraschall Med. 40(2):163-175, 2019 Apr.	Review/Other-Dx	N/A	To summarize the value of gastrointestinal ultrasound (GIUS) in the management of acute appendicitis and diverticulitis.	No results stated in abstract.	4
26. Etzioni DA, Mack TM, Beart RW, Jr., Kaiser AM. Diverticulitis in the United States: 1998-2005: changing patterns of disease and treatment. Ann Surg 2009;249:210-7.	Review/Other-Dx	267,000 patients	To analyze that the treatment of diverticulitis evolved over time and are reflected in patterns of practice in a nationally-representative patient cohort	Overall annual age-adjusted admissions for acute diverticulitis increased from 120,500 in 1998 to 151,900 in 2005 (26% increase). Rates of admission increased more rapidly within patients aged 18 to 44 years (82%) and 45 to 74 years (36%). Elective operations for diverticulitis rose from 16,100 to 22,500 per year during the same time period (29%), also with a more rapid increase (73%) in rates of surgery for individuals aged 18 to 44 years. Multivariate analysis found no evidence that primary anastomosis is becoming more commonly used.	4

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27. Andeweg CS, Knobben L, Hendriks JC, Bleichrodt RP, van Goor H. How to diagnose acute left-sided colonic diverticulitis: proposal for a clinical scoring system. <i>Ann Surg</i> . 2011;253(5):940-946.	Observational-Dx	124 patients with ALCD; 163 patients with other diagnoses	To assess and compare the diagnostic value of elements of the disease history, physical examination, and routine laboratory tests in patients with suspected ALCD.	Of 1,290 patients with acute abdominal pain, 287 patients were eligible for analysis. ALCD was the final diagnosis in 124 patients (43%). Age, 1 or more previous episodes, localization of symptoms in the lower left abdomen, aggravation of pain on movement, the absence of vomiting, localization of abdominal tenderness in the lower left abdomen, and C-reactive protein 50 or more were found to be independent predictors of ALCD. A nomogram was constructed based on these independent predictors with a diagnostic accuracy of 86%.	3
28. Tursi A, Brandimarte G, Di Mario F, et al. Predictive value of the Diverticular Inflammation and Complication Assessment (DICA) endoscopic classification on the outcome of diverticular disease of the colon: An international study. <i>United European Gastroenterol J</i> 2016;4:604-13.	Observational-Dx	1651 patients	To assess retrospectively the predictive value of DICA in patients for whom endoscopic data and clinical follow-up were available.	We enrolled 1651 patients (793 M, 858 F, mean age 66.6 +/- 11.1 years): 939 (56.9%) patients were classified as DICA 1, 501 (30.3%) patients as DICA 2 and 211 (12.8%) patients as DICA 3. The median follow-up was 24 (9-38) months. Acute diverticulitis (AD) occurred/recurred in 263 (15.9%) patients; surgery was necessary in 57 (21.7%) cases. DICA was the only factor significantly associated to the occurrence/recurrence of diverticulitis and surgery either at univariate ($\chi^2 = 405.029$; $p < 0.0001$) or multivariate analysis (hazard ratio = 4.319, 95% confidence interval (CI) 3.639-5.126; $p < 0.0001$). Only in DICA 2 patients was therapy effective for prevention of AD occurrence/recurrence with a hazard ratio (95% CI) of 0.598 (0.391-0.914) ($p = 0.006$, log rank test). Mesalazine-based therapies reduced the risk of AD occurrence/recurrence and needs of surgery with a hazard ratio (95% CI) of 0.2103 (0.122-0.364) and 0.459 (0.258-0.818), respectively.	3

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29. La Torre M, Mingoli A, Brachini G, et al. Differences between computed tomography and surgical findings in acute complicated diverticulitis. <i>ASIAN J. SURG.</i> 43(3):476-481, 2020 Mar.	Observational-Dx	152 patients	To study the inter-observer agreement of CT scan data and their concordance with the preoperative clinical findings and the adherence with the intraoperative status using a new classification of diverticular disease (CDD).	Overall inter-observer agreement for the CDD was high, with a kappa value of 0.905 (95% CI = 0.850-0.960) for observers A and B, while the concordance between radiological and surgical findings was weak (kappa values = 0.213 and 0.248, respectively and 95% CI = 0.106 to 0.319 and 95% CI = 0.142 to 0.355, respectively). When overall morbidity, mortality and the need of a terminal colostomy were considered as main endpoints no concordance was observed between surgical and radiological findings and the CDD (P=NS).	3
30. Bates DDB, Fernandez MB, Ponchiardi C, et al. Surgical management in acute diverticulitis and its association with multi-detector CT, modified Hinchey classification, and clinical parameters. <i>Abdominal Radiology</i> . 43(8):2060-2065, 2018 08. <i>Abdom Radiol</i> . 43(8):2060-2065, 2018 08.	Observational-Dx	301 patients	To assess whether CT findings, clinical parameters, and modified Hinchey classification are predictive of management outcomes of patients with acute diverticulitis.	CT findings associated with surgical management include the presence of a pericolic fluid collection (36.8% vs. 8.2%, P = 0.0001), colonic fistula (13.2% vs. 0.4%, P = 0.0002), extraluminal air (26.4% vs. 9.3%, P = 0.0052), bowel obstruction (5.2% vs. 0.0%, P = 0.0162), and a modified Hinchey score of Ib or higher (55.3% vs. 11.7%, P = 0.0001). Slightly lower systolic blood pressure was also associated with operative management (137.2 mmHg vs. 128.2 mmHg, P = 0.0220).	3

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31. Ebersole J, Medvecz AJ, Connolly C, et al. Comparison of American Association for the Surgery of Trauma grading scale with modified Hinchey classification in acute colonic diverticulitis: A pilot study. J Trauma Acute Care Surg. 88(6):770-775, 2020 06.	Observational-Dx	129 patients	To compare the AAST grading scale for acute colonic diverticulitis with the traditional Hinchey classification.	Of the total patients, 42.6% required procedural intervention, 21.7% required ICU admission, 18.6% were readmitted, and 6.2% died. Both AAST and Hinchey predicted the need for operation (AAST odds ratios, 1.55, 12.7, 18.09, and 77.24 for stages 2-5; Hinchey odds ratios, 8.85, 11.49, and 22.9 for stages 1b-3, stage 4 predicted perfectly). The need for operation c-statistics (area under the curve) for AAST and Hinchey was 0.80 and 0.83 for Hinchey and AAST, respectively (p = 0.35). The complication c-statistics curve for AAST and Hinchey was 0.83 and 0.80, respectively (p = 0.33). The AAST and Hinchey scores were less predictive for ICU admission, readmission, and mortality with c-statistics of less than 0.80.	4
32. Kim DH, Kim HJ, Jang SK, Yeon JW, Shin KS. CT Predictors of Unfavorable Clinical Outcomes of Acute Right Colonic Diverticulitis. AJR Am J Roentgenol. 209(6):1263-1271, 2017 Dec.	Observational-Dx	394 patients	To retrospectively assess CT predictors of unfavorable outcomes of medical treatment in patients with right colonic diverticulitis.	Of the 328 patients, nine underwent surgery after failed medical treatment. Of the other 319 patients, 35 had recurrence and 49 had a prolonged hospital stay. The spilled feces sign (adjusted odds ratio [OR], 111; p < 0.001) and serum WBC count (adjusted OR, 1.3; p = 0.047) were independent predictors of the need for surgery. More than five multiple diverticula per 10 cm of colon was significantly associated with recurrence (adjusted hazard ratio, 4.1; p < 0.001). Abscess larger than 4 cm (adjusted OR, 18.2; p = 0.01) and inflamed diverticulum larger than 2 cm (adjusted OR, 3.7; p = 0.001) were independent predictors of prolonged hospital stay.	3

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33. Buchs NC, Konrad-Mugnier B, Jannot AS, Poletti PA, Ambrosetti P, Gervaz P. Assessment of recurrence and complications following uncomplicated diverticulitis. Br J Surg. 100(7):976-9; discussion 979, 2013 Jun.	Observational-Tx	280 patients	To assess the risk of a recurrent attack following the first episode of uncomplicated diverticulitis.	During a median follow-up of 24 (range 3-63) months, 46 (16.4 per cent) of 280 patients experienced a second episode of diverticulitis. Six patients (2.1 per cent) subsequently developed complicated diverticulitis and four (1.4 per cent) underwent emergency surgery for peritonitis. In multivariable analysis, a raised serum level of C-reactive protein (over 240 mg/l) during the first attack was associated with early recurrence (hazard ratio 1.75, 95 per cent confidence interval 1.04 to 2.94; P = 0.035).	2
34. Gielens MP, Mulder IM, van der Harst E, et al. Preoperative staging of perforated diverticulitis by computed tomography scanning. Tech Coloproctol. 2012;16(5):363-368.	Observational-Dx	75 patients	To assess the accuracy of preoperative staging of perforated diverticulitis by CT scanning.	75 patients were included, 48 of whom (64%) were classified Hinchey 3 or 4 perforated diverticulitis during surgery. The PPV of preoperative CT scanning for different stages of perforated diverticulitis ranged from 45%–89%, and accuracy was between 71%–92%. The combination of a large amount of free intra-abdominal air and fluid was strongly associated with Hinchey 3 or 4 and therefore represented a reliable indicator for required surgical treatment.	2
35. Hill BC, Johnson SC, Owens EK, Gerber JL, Senagore AJ. CT scan for suspected acute abdominal process: impact of combinations of IV, oral, and rectal contrast. World J Surg. 2010;34(4):699-703.	Observational-Dx	661 patients	To evaluate the diagnostic accuracy of abdominal/pelvic CT with varying use of contrast agents in hospitalized patients.	A total of 661 patients were identified. Use of IV contrast alone was found in 54.2% of CT scans and was correct in 92.5% of cases. IV and oral contrast was used in 22.2% of CT scans and was 94.6% correct. Unenhanced imaging was performed in 16.2% and was correct in 92.5%. Oral contrast alone was used in 7.0% and was 93.5% correct. There was no significant difference in the ability to correctly diagnose a suspected acute abdominal process when enhanced CT imaging was compared to unenhanced (P>0.05).	3

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36. Rao PM, Rhea JT, Novelline RA, et al. Helical CT with only colonic contrast material for diagnosing diverticulitis: prospective evaluation of 150 patients. AJR Am J Roentgenol. 1998;170(6):1445-1449.	Observational-Dx	150 patients	Prospective study to determine sensitivity, specificity, and predictive value of CT for diverticulitis.	CT had a sensitivity of 97%, a specificity of 100%, a PPV of 100%, a NPV of 98%, and an overall accuracy of 99% for detection of diverticulitis.	3
37. Laqmani A, Veldhoen S, Dulz S, et al. Reduced-dose abdominopelvic CT using hybrid iterative reconstruction in suspected left-sided colonic diverticulitis. Eur Radiol. 26(1):216-24, 2016 Jan.	Observational-Dx	25 patients	To assess the effect of hybrid iterative reconstruction (HIR) and filtered back projection (FBP) on abdominopelvic CT with reduced-dose (RD-APCT) in the evaluation of acute left-sided colonic diverticulitis (ALCD).	OIN was reduced up to 54 % with HIR compared to FBP. Subjective image quality of HIR images was superior to FBP; subjective image noise was reduced. The detection rate of extraluminal air was higher with HIR L6. Reviewer confidence in interpreting CT findings of ALCD significantly improved with application of HIR.	3
38. Walter SS, Maurer M, Storz C, et al. Effects of Radiation Dose Reduction on Diagnostic Accuracy of Abdominal CT in Young Adults with Suspected Acute Diverticulitis: A Retrospective Intraindividual Analysis. Acad Radiol. 26(6):782-790, 2019 06.	Observational-Dx	44 patients	To assess the effects of radiation dose reduction on image quality and diagnostic accuracy of abdominal computed tomography (CT) in young adults with suspected acute diverticulitis.	Among 54 patients (mean age: 35.2 +/- 5.3 years, 77.8% male), the prevalence of acute diverticulitis was high (57.4%). Subjective image quality was highest for original datasets and lowest for LDCT datasets with 25% of the original dose (median [interquartile range]: 5 [5] vs. 3 [2-3], $p < 0.001$). Diagnostic confidence was high for all datasets down to 50% of the original dose, while 25% LDCT datasets were associated with a significantly decreased diagnostic confidence ($p < 0.001$). Diagnostic accuracy was high for all LDCT and original datasets (sensitivity: 100%, negative predictive value [NPV]: 100% for 75% and 100% dose levels; sensitivity: 96.8%, NPV: 95.8% for 50% dose level; sensitivity: 93.6%, NPV: 91.7% for 25% dose level, respectively). Inter-rater agreement regarding the detection of diverticulitis was almost perfect at doses $\geq 50\%$ (kappa: >0.81), while lower for datasets of 25% of the original radiation dose agreement (kappa: 0.67-0.78).	3

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EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
39. Dickerson EC, Chong ST, Ellis JH, et al. Recurrence of Colonic Diverticulitis: Identifying Predictive CT Findings-Retrospective Cohort Study. Radiology. 285(3):850-858, 2017 12.	Observational-Dx	440 patients	To identify computed tomographic (CT) findings that are predictive of recurrence of colonic diverticulitis.	Colonic diverticulitis most commonly involved the rectosigmoid (70%, 309 of 440) and descending (30%, 133 of 440) colon segments. Complicated diverticulitis was present in 22% (98 of 440) of patients. On the basis of the results of univariate analysis, significant predictors of diverticulitis recurrence were determined to be maximum colonic wall thickness in the inflamed segment (hazard ratio [HR], 1.07 per every millimeter of increase in wall thickness; $P < .001$), presence of a complication (HR, 1.75; $P = .002$), and subjective severity of inflammation (HR, 1.36 for every increase in severity category; P value for linear trend = .003). The difference in maximum wall thickness in the inflamed segment (HR, 1.05 per millimeter; $P = .016$) and subjective inflammation severity (HR, 1.29 per category; $P = .018$) remained statistically significant in a Cox multiple regression model.	3
40. Hinchey EJ, Schaal PG, Richards GK. Treatment of perforated diverticular disease of the colon. Adv Surg 1978;12:85-109.	Review/Other-Dx	N/A	To discuss the treatment of perforated diverticular disease of the colon.	No results stated in abstract.	4
41. Schreyer A, Layer G. German Society of Digestive and Metabolic Diseases (DGVS) as well as the German Society of General and Visceral Surgery (DGAV) in collaboration with the German Radiology Society (DRG). S2k guidelines for diverticular disease and diverticulitis: diagnosis, classification, and therapy for the radiologist. Rofo 2015;187:676-84.	Review/Other-Dx	N/A	To present guidelines for diverticular disease and diverticulitis by discussing the diagnosis, classification and therapy.	No results stated in abstract.	4

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EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
42. Flor N, Maconi G, Sardanelli F, et al. Prognostic Value of the Diverticular Disease Severity Score Based on CT Colonography: Follow-up in Patients Recovering from Acute Diverticulitis. Acad Radiol. 22(12):1503-9, 2015 Dec.	Observational-Dx	252 patients	To assess the prognostic value of a diverticular disease severity score (DDSS) based on computed tomography colonography (CTC) after acute diverticulitis (AD).	Significant correlation was found between CTC-based DDSS and clinical follow-up ($P = 0.022$) or elective surgery ($P = 0.007$), but not between clinical follow-up and CT-based score, extraluminal gas, C-reactive protein serum level, age, gender, or first versus recurrent AD episode. CTC demonstrated relevant additional findings in five of 46 (11%) patients: two AD complications (enterocolic and enterotubal fistulae), two colon cancers, and one extracolonic (lung) cancer.	4
43. Jerjen F, Zaidi T, Chan S, et al. Magnetic Resonance Imaging for the diagnosis and management of acute colonic diverticulitis: a review of current and future use. J Med Radiat Sci 2021;68:310-19.	Meta-analysis	13 studies	To review current and future use of MRI for the diagnosis and management of acute colonic diverticulitis.	Several existing MRI protocols are deemed suitable for ACD imaging, and it is recommended they be re-evaluated in larger cohorts. Future studies should consider the rapidly growing technological improvements of MRI, its cost efficiency and its applicability in modern day healthcare settings when addressing ACD management.	Good

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EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
44. Hall JF, Roberts PL, Ricciardi R, et al. Long-term follow-up after an initial episode of diverticulitis: what are the predictors of recurrence? Dis Colon Rectum 2011;54:283-8.	Observational-Dx	954 patients	To determine the clinical and CT predictors of recurrent disease after a first episode of diverticulitis that was successfully managed nonoperatively.	The study population included 672 patients; mean age, 61 +/- 15 years; mean follow-up, 42.8 +/- 24 months. The index presentation of diverticulitis was most commonly located in the sigmoid colon (72%), followed by descending colon (33%), right colon (5%), and transverse colon (3%). Overall recurrence at 5 years was 36% by (95% CI 31.4%-40.6%) Kaplan-Meier estimate. Complicated recurrence (fistula, abscess, free perforation) occurred in 3.9% (95% CI 2.2%-5.6%) of patients at 5 years by Kaplan-Meier estimate. Family history of diverticulitis (HR 2.2, 95% CI 1.4-3.2), length of involved colon >5 cm (HR 1.7, 95% CI 1.3-2.3), and retroperitoneal abscess (HR 4.5, 95% CI 1.1-18.4) were associated with diverticulitis recurrence. Right colon disease (HR 0.27, 95% CI 0.09-0.86) was associated with freedom from recurrence.	3
45. Poletti PA, Platon A, Rutschmann O, et al. Acute left colonic diverticulitis: can CT findings be used to predict recurrence? AJR Am J Roentgenol 2004;182:1159-65.	Observational-Dx	312 patients	To explore CT and demographic predictors for unfavorable outcome of nonoperative treatment in patients with a first event of left colonic diverticulitis.	Among these 168 patients, 115 (68%) had an uneventful outcome, but nonoperative treatment failed in 53 (32%). The presence of an abscess (n = 19) or extraintestinal gas pocket (n = 14) were the only CT findings significantly associated with failure of nonoperative treatment. Adjusted odds ratios (95% confidence interval) for failure were 6.18 (1.76-21.68) when an abscess was diagnosed and 4.26 (1.04-17.57) when pockets of free air were observed. Sex and age were not significantly associated with unfavorable outcome of nonoperative treatment.	2

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EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
46. Ambrosetti P, Becker C, Terrier F. Colonic diverticulitis: impact of imaging on surgical management -- a prospective study of 542 patients. Eur Radiol 2002;12:1145-9.	Observational-Dx	542 patients	To compare the performance of the CT and the water-soluble contrast enema (CE) in the diagnosis and the severity of acute left-colonic diverticulitis, and to recognize the impact of CT during the acute phase and after a first acute episode successfully treated medically.	The performance of CT is significantly superior to CE in terms of sensitivity (98 vs 92%, $p<0.01$), and in the evaluation of the severity of the inflammation (26 vs 9%, $p<0.02$). Moreover, of 69 patients who had an associated abscess seen on CT, only 20 (29%) had indirect signs of this complication on CE. During the acute phase the chances of medical treatment failure are statistically greater when diverticulitis is considered severe on CT than when it is considered moderate (26% for the severe diverticulitis vs 4% for the moderate ones, $p<0.0001$). After successful medical treatment of the acute episode, patients with severe diverticulitis on the CT had statistically greater incidence of secondary bad outcome than patients with moderate diverticulitis (36 vs 17%, $p<0.0001$).	3
47. Najjar SF, Jamal MK, Savas JF, Miller TA. The spectrum of colovesical fistula and diagnostic paradigm. Am J Surg. 188(5):617-21, 2004 Nov.	Review/Other-Dx	12 patients	To evaluate this condition over a 12-year period with special emphasis on its clinical presentation, etiologic factors involved, and modalities used to verify its diagnosis.	Underlying etiologies were diverticular disease (75%), colon cancer (16%), and bladder cancer (8%). Pneumaturia (77%) was the most common presentation, followed by urinary tract infections, dysuria and frequency (45%), fecaluria (36%), hematuria (22%), and orchitis (10%). The ability of various preoperative investigations to identify a CVF were: computed tomography (CT) (90%), barium enema (BE) (20%), and cystography (11%), whereas cystoscopy, intravenous pyelogram (IVP), and colonoscopy were nondiagnostic. All patients underwent single- or multiple-staged repair of the fistula.	4

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EVIDENCE TABLE

Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
48. Miyaso H, Iwakawa K, Hamada Y, et al. Ten Cases of Colovesical Fistula due to Sigmoid Diverticulitis. Hiroshima Journal of Medical Sciences. 64(1-2):9-13, 2015 Jun. Hiroshima J Med Sci. 64(1-2):9-13, 2015 Jun.	Observational-Dx	10 patients	To review colovesical fistula due to sigmoid diverticulitis over a 9-year period to clarify presentation and diagnostic confirmation.	Preoperative urinalysis showing bacteriuria (100%) was the most common presentation, followed by fecaluria (40%), abdominal pain (40%), pneumaturia (30%), hematuria (30%), pain on urination (30%), pollakiuria (10%), and dysuria (10%). The abilities of various preoperative investigations to identify CVF were: computed tomography (CT), 88.9%; magnetic resonance imaging, 40%; cystoscopy, 30%, and gastrografin irrigoscopy, 22.2%. Colonoscopy (0%) was not diagnostic. Bowel resection was performed in nine of ten patients. When inflammation was intense, covering ileostomy was performed, and an omental plasty was placed between the bowel anastomosis and bladder.	4
49. Nielsen K, Richir MC, Stolk TT, et al. The limited role of ultrasound in the diagnostic process of colonic diverticulitis. World J Surg. 38(7):1814-8, 2014 Jul.	Observational-Dx	232 patients	To determine the diagnostic accuracy of US compared with CT for patients with uncomplicated and complicated diverticulitis.	A total of 123 patients underwent an US and a CT scan. In 78/94 patients with uncomplicated diverticulitis, results of US and CT scan were compatible (83 %); in 6 of the remaining 29 patients both modalities showed a complicated diverticulitis (21 %). US misdiagnosed 17 % of patients with uncomplicated diverticulitis and 79 % with complicated diverticulitis.	3
50. Macia-Suarez D. Is it useful ultrasonography as the first-line imaging technique in patients with suspected acute diverticulitis?. [Review]. RADIOLOGIA. 61(6):506-509, 2019 Nov - Dec.	Review/Other-Dx	N/A	To review the scientific evidence about the diagnostic validity of ultrasonography and computed tomography in the diagnosis of acute diverticulitis of the colon.	No results stated in abstract.	4

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Reference	Study Type	Patients/ Events	Study Objective (Purpose of Study)	Study Results	Study Quality
51. American College of Radiology. ACR Appropriateness Criteria® Radiation Dose Assessment Introduction. Available at: https://edge.sitecorecloud.io/americancoldf5f-acrorgf92a-productioncb02-3650/media/ACR/Files/Clinical/Appropriateness-Criteria/ACR-Appropriateness-Criteria-Radiation-Dose-Assessment-Introduction.pdf .	Review/Other-Dx	N/A	To provide evidence-based guidelines on exposure of patients to ionizing radiation.	No abstract available.	4

Evidence Table Key

Study Quality Category Definitions

- *Category 1* The study is well-designed and accounts for common biases.
- *Category 2* The study is moderately well-designed and accounts for most common biases.
- *Category 3* There are important study design limitations.
- *Category 4* The study is not useful as primary evidence. The article may not be a clinical study or the study design is invalid, or conclusions are based on expert consensus. For example:
 - a. The study does not meet the criteria for or is not a hypothesis-based clinical study (e.g., a book chapter or case report or case series description);
 - b. The study may synthesize and draw conclusions about several studies such as a literature review article or book chapter but is not primary evidence;
 - c. The study is an expert opinion or consensus document.
- Meta-analysis
 - a. *Good quality* – the study design, methods, analysis, and results are valid and the conclusion is supported.
 - b. *Inadequate quality* – the study design, analysis, and results lack the methodological rigor to be considered a good meta-analysis study.

Abbreviations Key

Dx = Diagnostic
Tx = Treatment